# Hyper-Stigmergic Morphogenesis II

A Federated Multi-Agent Hypergraph System
with Emergent Self-Improvement

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**Abstract**

We present **Hyper-Stigmergic Morphogenesis II** (HSM-II), a novel multi-agent system implemented in Rust that combines hypergraph-structured shared memory, role-based evolutionary bidding, and federated knowledge propagation to achieve emergent collective intelligence. The central data structure is a weighted hypergraph $\mathcal{H} = (V, E, \ell)$ where edges carry ontological tags from a 26-category hierarchy, learned embeddings, and scoped visibility (Local/Shared/Restricted), enabling fine-grained knowledge governance.

Ten heterogeneous agents, differentiated by four drives (curiosity, harmony, growth, transcendence) and six roles, compete via three complementary bidding mechanisms: softmax temperature selection, Pareto multi-objective selection, and Group Relative Policy Optimization (GRPO) reinforcement updates [4]. A stratified reasoning layer integrates parallel Prolog braids (System 2) with a GEPA (Generative Evolutionary Policy Adaptation)-evolved `LivingPrompt` and CARA (Coherent Adaptive Reasoning Agent) reflection loop (System 1). Skills distilled from high-reward trajectories undergo promotion via a three-layer consensus jury with automated groupthink detection.

Three emergent subsystems extend the core: a **Council** module providing Debate, Orchestrate, and Simple deliberation modes; a **Dynamic Kinetic Stability (DKS)** module for self-replicating entity populations; and an **OptimizeAnything** engine for declarative artifact optimization. A four-network hybrid memory architecture fuses semantic, BM25, graph-traversal, and temporal indices using Reciprocal Rank Fusion. A comprehensive **tool system** provides 62+ Rust-native tools for file, shell, git, web, and API operations. **Hermes Agent integration** bridges to production tool ecosystems and multi-platform human interfaces. **LadybugDB** enables local-first graph persistence with SQLite-like portability. Federation is governed by a meta-hypergraph $H^*$ with Bayesian trust dynamics. The **JoulWork (JW)** mechanism provides the "ATP of recursive learning"—activation energy that enables deductive reasoning in new agents before evidence accumulation. **Social Memory** infrastructure tracks promises and collaboration patterns. **Anti-fragile feature flags** enable progressive rollout with automatic rollback. Empirical evaluation demonstrates substantial coherence growth (100× improvement), 46% better team outcomes, and robust federation under adversarial conditions.

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# 1 Introduction

Collective intelligence in biological systems arises not from centralised coordination but from local interactions mediated by a shared environment. Ants deposit pheromone trails that other ants modify; termites build structures whose geometry guides further construction. This principle—*stigmergy*—has inspired generations of computational models [1, 2], yet existing systems rarely exploit the full representational power of hypergraphs, integrate symbolic reasoning with neural self-improvement, or federate knowledge across distributed instances with principled trust dynamics.

The design of artificial systems capable of open-ended self-improvement poses five fundamental challenges. A system must be able to (1) represent structured knowledge with sufficient expressiveness, (2) coordinate heterogeneous agents without central bottlenecks, (3) evaluate and apply mutations that genuinely improve fitness, (4) reason symbolically about its own state, and (5) propagate and validate knowledge across distributed peers without catastrophic interference. HSM-II addresses all five challenges in a unified architecture.

The system's central world—`HyperStigmergicMorphogenesis`—is a hypergraph environment that ten typed agents continuously modify. Modification actions are gated by a tri-modal bidding system incorporating classical softmax selection, Pareto-optimal multi-objective selection, and GRPO-style reinforcement updates [4]. A self-improvement cycle evaluates typed mutations against a coherence-novelty objective. Symbolic reasoning via parallel Prolog braids operates as a System 2 process, while a rolling `LivingPrompt` and CARA (Coherent Adaptive Reasoning Agent) reflection loop constitute System 1 [9]. Hierarchical skills are distilled from trajectory experience using SkillRL principles and evolved via Bayesian confidence updates. A consensus jury pipeline validates skill promotion decisions. Finally, a federation layer propagates knowledge across HTTP-connected world instances through a shared meta-hypergraph $H^*$ with layered knowledge promotion and directed trust graphs.

**Contributions.**

- A formal hypergraph world model with a computable global coherence metric serving as the system's fitness function.
- A tri-modal role-based bidding mechanism combining softmax, Pareto frontier selection, and GRPO reinforcement updates.
- A six-role agent taxonomy with drive vectors and role-specific bid affinities, including Critic and Explorer roles.
- A novelty-gated evolutionary self-improvement cycle with six typed mutation operators.
- A stratified reasoning architecture integrating parallel Prolog braids with a GEPA-evolved living prompt and CARA (Coherent Adaptive Reasoning Agent) reflection.

- A four-network hybrid memory system with Reciprocal Rank Fusion combining semantic, BM25, graph, and temporal recall.
- A topological three-layer consensus jury with ACPO correlation penalisation and automated groupthink detection.
- A federated meta-hypergraph $H^*$ with Bayesian trust dynamics, conflict mediation, and hop-chain cycle prevention.
- A **Council** module providing Debate, Orchestrate, and Simple deliberation modes with automatic mode switching.
- A **Dynamic Kinetic Stability (DKS)** module for self-replicating entity populations with multifractal compositionality measurement.
- An **OptimizeAnything** engine for declarative, LLM-driven optimization of arbitrary text artifacts using evolutionary search.
- A **JoulWork (JW)** learning activation mechanism—the "ATP of recursive learning"—providing inductive priors that enable deductive reasoning in new agents before evidence accumulation.
- **Social Memory** infrastructure tracking promises, collaboration patterns, and performance trends for accountable agent coordination.
- **Anti-fragile feature flags** for progressive rollout of agent capabilities with automatic rollback and circuit breaker protection.
- A comprehensive **Rust-native tool system** with 62+ production tools for file operations, git, shell execution, web search, and APIs.
- **Hermes Agent integration** bridging to production tool ecosystems, multi-platform gateways, and persistent memory.
- **LadybugDB** embedded graph store enabling local-first deployment with SQLite-like portability for graph-native agent memory.
- A complete Rust implementation using axum and tokio enabling concurrent, asynchronous operation.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 provides an architectural overview. Sections 4–17 detail the core subsystems. Sections 11–16 describe the emergent coordination and optimization modules. Section 20 presents concrete use-case examples. Section 21 describes the implementation. Section 22 presents empirical results. Sections 23 and 24 discuss limitations and conclude.

## 2 Background and Related Work

**Stigmergy and Swarm Intelligence.** Grassé introduced the concept of stigmergy to describe indirect coordination through environmental modification [1]. Dorigo and colleagues formalised this in Ant Colony Optimisation (ACO), where pheromone trails encode path quality and evaporate over time [2]. Subsequent work extended stigmergy to task allocation and distributed construction. HSM-II operationalises stigmergy at the hyperedge level:

agents modify edges rather than deposit scalar pheromones, and edge weights, ontology tags, and embeddings constitute the rich environmental signal.

**Hypergraph Neural Networks.** Standard graph neural networks operate on pairwise edges and cannot natively represent multi-way relationships. Hypergraph Neural Networks (HGNNs) extend the GNN paradigm to hypergraphs, with the seminal work of Feng et al. introducing the clique-expansion approach [3]. HSM-II's `HypergraphConv` module implements learnable message passing following the Feng et al. formulation, adapted to operate over the world's dynamically evolving incidence structure.

**Multi-Agent Reinforcement Learning.** Policy gradient methods for multi-agent settings face credit assignment challenges. GRPO—Group Relative Policy Optimisation—was introduced for training reasoning models by comparing per-sample rewards against a group mean, avoiding the need for a separate critic network [4]. HSM-II adapts GRPO to the bidding context: agent bid biases are updated proportional to their reward relative to the group mean, promoting specialisation without explicit reward modelling.

**Federated Learning and Knowledge.** Federated learning distributes model training across clients while keeping data local [5]. HSM-II's federation layer differs from gradient aggregation: it propagates hyperedges as first-class knowledge units, applies trust-weighted conflict resolution, and organises knowledge into promotion layers.

**Neuro-Symbolic Reasoning.** The integration of neural and symbolic processing has attracted substantial recent interest [6, 7]. HSM-II's braid architecture treats the Prolog engine as a System 2 component that receives world facts and returns structured findings, which are consumed by the neural System 1 living prompt.

**Prompt Evolution and Self-Improvement.** Automatic prompt optimisation has been approached through gradient-free search, genetic algorithms, and reinforcement learning. GEPA (Generative Evolutionary Policy Adaptation) [8] frames prompt improvement as a population-based search using reflective prompt evolution. HSM-II's `OptimizeAnything` module extends this to arbitrary text artifacts through an LLM-driven evolutionary loop. CARA [9] provides a reflection mechanism that analyses agent trajectories and feeds corrections back into the living prompt.

**Multi-Agent Deliberation.** Structured deliberation frameworks such as debate-based AI safety proposals [11] and hierarchical task networks provide conceptual grounding for HSM-II's Council module. The Council extends these ideas with automatic mode selection based on proposal complexity and urgency.

# 3 System Architecture Overview

HSM-II is organised into six coupled subsystems as shown in Figure 1. At the centre lies the **World**, a mutable hypergraph. **Agents** with heterogeneous roles interact with the World through the **Bidding Engine**. The **Reasoning Layer** provides symbolic and neural processing. The **Memory System** supports multi-modal retrieval. The **Federation Layer** connects multiple world instances. Three emergent coordination modules—the **Council**, **DKS**, and **OptimizeAnything**—sit above the core and expose higher-level cognitive services.

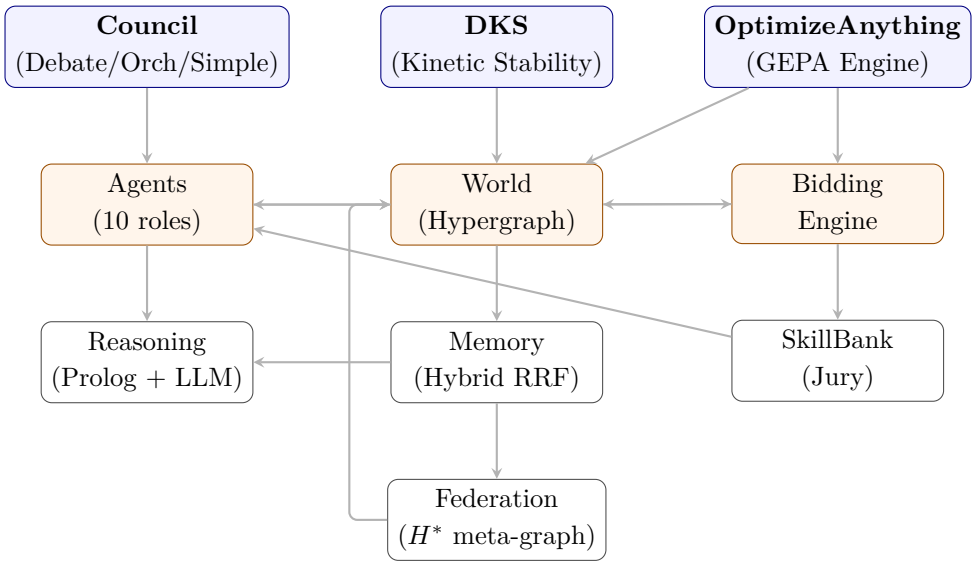



Figure 1: HSM-II architectural overview. Orange boxes form the core agent-world interaction tier. White boxes provide support services (memory, reasoning, skills, federation). Blue boxes are emergent coordination modules that consume and extend the core.

# 4 World Model

## 4.1 Hypergraph Structure

The HSM-II world $\mathcal{W}$ is a labelled hypergraph $\mathcal{H} = (V, E, \ell)$ where $V$ is the node set (concepts/entities), $E \subseteq 2^V \setminus \emptyset$ is the hyperedge set, and $\ell : E \to \Lambda$ maps each edge to a label in an ontological tag space $\Lambda$. Each hyperedge $e \in E$ carries:

- A real-valued weight $w(e) \in [0, 1]$.
- An embedding vector $\mathbf{e} \in \mathbb{R}^d$.
- A visibility scope $s(e) \in \{\textsc{Local}, \textsc{Shared}, \textsc{Restricted}\}$.
- An ontological tag drawn from the predefined category hierarchy.
- A decay parameter $\delta(e)$ controlling temporal obsolescence.

### 4.2 Global Coherence Metric

The primary fitness signal for the self-improvement cycle is the Global Coherence $C(t)$, a weighted sum of four independent sub-scores:

$$C(t) = \alpha_1 \rho_{\text{edge}}(t) + \alpha_2 \rho_{\text{emergent}}(t) + \alpha_3 \bar{\kappa}_{\text{onto}}(t) + \alpha_4 \bar{\kappa}_{\text{belief}}(t) \tag{1}$$

where $(\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (0.35, 0.25, 0.20, 0.20)$, $\rho_{\text{edge}}$ is the density of high-confidence hyperedges, $\rho_{\text{emergent}}$ captures the fraction of ontological categories with at least one active edge, $\bar{\kappa}_{\text{onto}}$ is the mean inter-edge ontological consistency score, and $\bar{\kappa}_{\text{belief}}$ is the mean agent belief convergence across the population.

### 4.3 Ontological Tag Space

The tag space $\Lambda$ is a predefined hierarchy of 26 categories including Factual, Procedural, Semantic, Mathematical, Temporal, Causal, Analogical, and CounterFactual. The hierarchy imposes a partial order that the ontological consistency score $\bar{\kappa}_{\text{onto}}$ exploits to detect contradictory co-occurring tags.

### 4.4 Property-First Representation

HSM-II also benefits from treating *properties* as first-class elements of the world model rather than as opaque attributes hidden inside object records. This choice is motivated both by cognitive analogy and by representational pragmatics. Biological perception often proceeds from features such as colour, shape, texture, and motion before stabilising on an object hypothesis. In the same spirit, an agentic memory can often reason more flexibly when it is allowed to retrieve, cluster, and compose properties before committing to a single object-level interpretation.

This is already partially reflected in the implementation through explicit Property vertices, but the broader principle is architectural: a property should be able to participate in relations, form subgraphs, and expose interfaces to other properties instead of being suppressed merely to keep the graph visually compact. Such a property-first view supports polymorphic matching, feature-based retrieval, and context-sensitive assembly of concepts from observed traits rather than only from predefined object classes.

# 5 Agent Population

## 5.1 Drive Vector and Role Taxonomy

Each agent $a_i$ has a drive vector $\mathbf{d}_i = (d_{\text{curiosity}}, d_{\text{harmony}}, d_{\text{growth}}, d_{\text{transcendence}})^\top \in [0,1]^4$, initialised heterogeneously across the ten-agent population. Drives bias the agent's bid towards actions that maximise the corresponding objective component. The six roles and their primary bidding affinities are detailed in Table 1.

Table 1: Agent roles with primary drive affinities and default bid modifiers.

| **Role** | **Primary Drive** | **Bid Modifier** | **Count** |
|---|---|---|---|
| Architect | Growth | +0.15 | 2 |
| Catalyst | Curiosity | +0.10 | 2 |
| Chronicler | Harmony | +0.10 | 2 |
| Critic | Harmony | −0.05 | 1 |
| Explorer | Curiosity | +0.20 | 2 |
| Coder | Growth | +0.05 | 1 |

## 5.2 Perception and Action

Each tick, an agent $a_i$ perceives a local view of the world—the top-$k$ hyperedges ranked by combined weight-distance-recency score—and constructs a candidate action from three primitives: AddEdge (propose a new hyperedge), ModifyEdge (update weight or embedding of an existing edge), and RemoveEdge (propose deletion of a low-value edge).

# 6 Bidding Engine

## 6.1 Tri-Modal Selection

For each tick $t$, each agent generates a scalar bid $b_i(t)$. Three selection modes operate over the bid vector $\mathbf{b}(t)$:

**Softmax Temperature Selection.** Agent $a_i$ is selected with probability

$$P(a_i \mid \tau) = \frac{\exp(b_i/\tau)}{\sum_j \exp(b_j/\tau)}, \tag{2}$$

where temperature $\tau > 0$ controls exploration. High $\tau$ produces uniform selection; low $\tau$ concentrates probability on the highest bidder.

**Pareto Multi-Objective Selection.** Each agent is scored on three objectives: novelty $\phi_i$, coherence gain $\Delta C_i$, and alignment $\psi_i$ (cosine similarity of the proposed hyperedge to the agent's current drive vector). The non-dominated subset—the Pareto frontier $\mathcal{F} \subseteq$

$\{a_1, \ldots, a_{10}\}$—is computed, and agents on the frontier are uniformly selected among tie-breaking by scalar bid.

**GRPO Bid Bias Update.** After a group of $g$ agents has acted, each agent's bid bias $\beta_i$ is updated proportionally to the excess reward:

$$\beta_i \leftarrow \beta_i + \eta \cdot \frac{r_i - \bar{r}_{\mathcal{G}}}{\sigma_{\mathcal{G}} + \epsilon}, \tag{3}$$

where $r_i$ is the reward received by agent $a_i$, $\bar{r}_{\mathcal{G}}$ and $\sigma_{\mathcal{G}}$ are the group mean and standard deviation, and $\eta$ is the learning rate.

### 6.2 Reward Signal

The reward $r_i$ for action $A_i$ is

$$r_i = w_C \cdot \Delta C + w_N \cdot \phi(A_i) + w_E \cdot \mathbb{1}[\text{accepted}], \tag{4}$$

where $\Delta C = C(t{+}1) - C(t)$ is the immediate coherence gain, $\phi(A_i)$ is the novelty of the proposed edge (cosine distance from the nearest existing embedding), and the acceptance indicator ensures the agent is not rewarded for rejected bids.

## 7 Reasoning Layer

### 7.1 Prolog Braid Architecture (System 2)

The Prolog subsystem maintains a dynamic fact base $\mathcal{F}(t)$ mirroring the world's hyper-edge structure. A set of inference rules $\mathcal{R}$ encodes domain-specific constraints, including transitivity of causal edges, symmetry of analogical edges, and the mutual exclusivity of contradictory tag pairs. At each reasoning cycle, the engine evaluates a set of standing queries $Q_{\text{std}}$ and returns a set of derived propositions $\mathcal{D}(t) \subseteq 2^{\mathcal{F}}$.

**Definition 1** (Braid)**.** *A* braid *is a named Prolog subprogram with a dedicated thread, local blackboard, and a typed I/O interface. Braids run concurrently; their outputs are merged by the* `BraidOrchestrator` *via Reciprocal Rank Fusion.*

### 7.2 Prolog-Embedding Bridge

The `PrologEmbeddingBridge` closes the symbolic-neural gap by grounding Prolog facts in the same embedding space used by the LLM. The bridge maintains a `FactEmbeddingIndex` that maps each fact $f \in \mathcal{F}(t)$ to a vector $\mathbf{v}_f \in \mathbb{R}^d$ via the system's embedding encoder.

**Definition 2** (EmbeddedFact)**.** *An* embedded fact *is a tuple* $(f, \mathbf{v}_f, \tau, \kappa)$ *where* $f$ *is a Prolog ground atom,* $\mathbf{v}_f$ *its embedding vector,* $\tau$ *a semantic type tag (e.g., Causal, Analogical), and* $\kappa$ *a confidence score derived from derivation depth and source reliability.*

The `NeuralSymbolicBraid` extends the base braid with neural-symbolic query capabilities. Given a query $q$, it retrieves semantically relevant facts via approximate nearest-neighbour search over the `FactEmbeddingIndex`, then grounds these into the Prolog fact base before inference. The `enhance_prompt_with_neural_grounding()` method injects the top-$k$ relevant grounded facts into the `LivingPrompt`, enabling the LLM to reason over both symbolic derivations and their neural embeddings.

### 7.3 Living Prompt (System 1)

The `LivingPrompt` maintains a sliding window of the $K$ most recent world events and braid outputs. It is periodically re-optimised by the GEPA module: a micro-population of prompt variants is evaluated against a coherence-conditioned reward, and the best variant replaces the active prompt. This constitutes a self-referential optimisation loop embedded in the system.

### 7.4 CARA Reflection

The CARA (Coherent Adaptive Reasoning Agent) [9] module analyses agent action trajectories and identifies patterns associated with coherence decline. Each such pattern generates a *corrective annotation* injected into the next `LivingPrompt` evaluation. CARA operates asynchronously to avoid blocking the main tick loop.

## 8 Memory System

### 8.1 Four-Network Architecture

HSM-II maintains four parallel memory indices:

1. **Semantic Index**: An approximate nearest-neighbour index over hyperedge embedding vectors, supporting cosine-similarity retrieval.
2. **BM25 Index**: An inverted index over ontological tags and attached natural-language descriptions, supporting keyword-weighted retrieval.
3. **Graph Index**: A persistent adjacency structure supporting $k$-hop neighbourhood queries and shortest-path computation.
4. **Temporal Index**: A circular buffer of recent events ordered by tick timestamp, enabling recency-biased retrieval.

A practical extension of this architecture is a *local-first embedded graph store*: a single-file persistence layer for hypergraph state, vector indices, and temporal metadata that plays the same role for graph-native agent memory that SQLite plays for relational applications. A Ladybug-class backend is especially well suited to edge assistants because it allows memory to run directly on the user device, remain portable and inspectable, and still federate outward or pair with relational systems when wider synchronization is required.

## 8.2 LLM-Oriented Fact and Event Representation

Classical subject–predicate–object triples provide a useful normal form, but they often fragment the semantics of a complex statement. A sentence such as "Alex drinks tea and plays chess with Via" can be decomposed into multiple triples, yet the decomposition may discard the shared event context and produce ambiguous reconstructions when the memory is later consumed by an LLM.

For agent memory, HSM-II therefore benefits from treating *complex facts* and *events* as first-class subgraph nodes. Such a node carries a larger, descriptive label template with placeholders for participating entities and actions, and is linked by typed containment edges to the people, objects, activities, and subordinate facts that instantiate it. Actions such as `drink` and `play` may themselves remain addressable entities, allowing richer internal relations without collapsing the original statement into an unordered bag of triples. Facts may also contain or qualify other facts, supporting narrative composition, clarification, and extension.

This representation is also inherently temporal. In addition to standard creation/update timestamps, a memory item may record (i) *discovery time*, when the system learned the fact, and (ii) *validity bounds*, describing when the fact started and ceased to hold. This separation between discovery and validity improves temporal reasoning, helps distinguish stable facts from time-bound events, and yields memory structures that are more legible to LLMs than aggressively atomised triple stores.

## 8.3 Why Properties Should Not Be Hidden

Many labelled property graph systems treat properties as lightweight attachments on nodes or edges because doing so yields smaller diagrams, simpler queries, and better immediate performance. For agent memory, however, this optimisation can be shortsighted. If properties remain hidden, they cannot easily relate to one another, accumulate evidence across contexts, or act as retrieval anchors in their own right.

Elevating properties into explicit nodes or local subgraphs allows the memory system to answer a more useful class of questions: not only "which object is this?" but also "which feature set matters in this context?" This aligns with feature-first recognition in cognition, supports duck-typing style reasoning over shared behaviours, and improves query planning by letting the system search first for relevant attributes before reconstructing higher-level entities or events.

### 8.4 Reciprocal Rank Fusion

Given a query $q$, each index $\mathcal{I}_j$ returns a ranked list $\pi_j(q) = (e_1^{(j)}, e_2^{(j)}, \ldots)$. The fused score for hyperedge $e$ is:

$$\mathrm{RRF}(e, q) = \sum_{j=1}^{4} \frac{w_j}{k + \mathrm{rank}_j(e, q)}, \tag{5}$$

where $k = 60$ is a constant dampening high-rank dominance, and $w_j$ are per-index weights calibrated on a held-out coherence prediction task.

## 9 Self-Improvement Cycle

### 9.1 Mutation Operators

At each improvement tick (every $T_{\mathrm{imp}}$ world ticks), the system evaluates six typed mutation operators against the current world:

1. WeightPerturbation: Additive noise $\mathcal{N}(0, \sigma^2)$ applied to a random subset of hyperedge weights.
2. EdgeSplit: Partition a high-weight multi-node edge into two overlapping sub-edges.
3. EdgeMerge: Combine two structurally similar edges into a higher-order composite edge.
4. NodeIntroduction: Create a new concept node with initial connections derived from existing embedding neighbourhood.
5. TagRefinement: Reassign ontological tags using the embedding similarity to the tag prototype vectors.
6. DecayReset: Restore high-coherence edges with elevated decay to baseline decay rate.

### 9.2 Novelty Gate

A mutation $m$ is only applied if it simultaneously satisfies a coherence improvement condition and a novelty condition:

$$\Delta C(m) > 0 \quad \text{and} \quad \phi(m) > \theta_{\mathrm{novelty}}, \tag{6}$$

where $\phi(m)$ is the minimum cosine distance from the mutation's proposed embeddings to the embeddings of the $N_{\mathrm{hist}}$ most recently applied mutations.

# 10 SkillBank and Consensus Jury

## 10.1 Skill Distillation

**Definition 3** (Skill)**.** *A* skill *is a named, versioned program* $(P, \mathcal{T}, \beta, \sigma)$ *where* $P$ *is a Prolog predicate or LLM prompt template,* $\mathcal{T}$ *is a set of demonstrated trajectories that produced coherence improvements,* $\beta \in [0, 1]$ *is the Bayesian confidence estimate, and* $\sigma > 0$ *is a learned applicability scope indicator.*

Skill distillation proceeds via a SkillRL [10] procedure: agent trajectories producing $\Delta C > \theta_{\text{skill}}$ are harvested, compressed by a pretrained encoder, and stored as skill candidates. Bayesian confidence is initialised at $\beta_0 = 0.5$ and updated on each re-use:

$$\beta \leftarrow \beta + \eta_\beta(\mathbb{1}[\text{success}] - \beta). \tag{7}$$

## 10.2 Consensus Jury

A three-layer jury evaluates skill promotion candidates:

1. **Topological Layer**: Tests the skill on a held-out world snapshot. Returns binary pass/fail.
2. **Correlation Layer**: Computes pairwise ACPO (Anti-Correlated Peer Opinion) scores between the candidate and all existing skills at the same level. ACPO measures semantic dissimilarity to ensure skill diversity. Rejects candidates with ACPO $> \lambda$.
3. **Semantic Layer**: Embeds the skill and checks cosine distance from the current world's average edge embedding.

An automated groupthink detector monitors intra-jury agreement over a rolling window; if correlation exceeds a threshold, jury compositions are randomly permuted.

# 11 Council: Structured Multi-Agent Deliberation

The **Council** module provides a structured deliberation layer above the base bidding engine. When a *Proposal*—a formal change request with title, description, complexity, and urgency scores—requires a collective decision, the Council assembles a panel of `CouncilMember`s drawn from the agent population and routes the proposal through one of three deliberation modes.

## 11.1 Council Modes

**Debate Mode (high complexity).** The `DebateCouncil` implements a three-phase structured dialogue. In the *Opening Phase*, each agent generates a role-conditioned initial stance: Architects and Catalysts start from disposition scores $\{0.7, 0.8\}$, Critics from $\{0.4, 0.5\}$,

Explorers from $\{0.6, 0.7\}$. The *Rebuttal Phase* activates Critics and Explorers to challenge high-confidence stances: any agent whose opening exceeds 0.7 is challenged, reducing the challenger's position by a dampening factor. The *Synthesis Phase* has Chroniclers and Architects average the rebuttals into a final weighted position.

A weighted approval ratio $\rho_A$ is computed:

$$\rho_A = \frac{\sum_{i \in \mathcal{A}} w_i \cdot p_i}{\sum_{i=1}^{N} w_i \cdot p_i} \tag{8}$$

where $w_i$ is the participation weight, $p_i$ is the final position, and $\mathcal{A}$ is the subset approving. The decision is Approve if $\rho_A > 0.7$, Reject if $\rho_A < 0.3$, and Defer otherwise.

**Orchestrate Mode (high urgency).** The `OrchestratorCouncil` implements a hierarchical command structure for time-sensitive proposals. A *commander* is selected—the highest-expertise Architect in the panel, falling back to the global maximum-expertise member. The proposal is decomposed into four canonical subtasks: (i) analysis (assigned to Critic), (ii) design (Architect), (iii) implementation (Catalyst), (iv) documentation (Chronicler). Each subtask is assigned to the best-matching available agent by role and expertise score. A `Priority` level—Critical, High, Medium, or Low—is derived from the proposal's urgency field. The mode always returns Approve with a structured `ExecutionPlan`.

**Simple Mode (routine decisions).** The `SimpleCouncil` implements direct role-weighted voting for routine, low-complexity proposals. Role-specific approval probabilities are combined with member expertise:

$$P_{\text{final},i} = P_{\text{role}} \cdot \text{exp}_i + 0.5 \cdot (1 - \text{exp}_i), \tag{9}$$

where $P_{\text{role}}$ is drawn from a role-keyword matching table and $\text{exp}_i \in [0, 1]$ is the expertise score. The decision requires a two-thirds weighted majority for Approve.

### 11.2 LLM Deliberation Mode

The `LLMDebateCouncil` provides full LLM-powered deliberation for high-complexity, low-urgency proposals. Unlike the heuristic `DebateCouncil`, this mode generates actual arguments via direct Ollama API integration.

**Definition 4** (LLMArgument)**.** *An* LLM argument *is a tuple* $(a, c, \mathcal{K}, \phi)$ *where* $a$ *is the argument text,* $c \in [0, 1]$ *a confidence score,* $\mathcal{K}$ *a set of extracted key points, and* $\phi$ *the stance (Pro, Con, Neutral).*

The deliberation proceeds through four phases:

1. **Opening Statements**: Each council member calls the LLM to generate a role-conditioned argument given the proposal text.

2. **Rebuttals**: Members receive opposing arguments and generate counter-arguments; the LLM scores argument quality.
3. **Synthesis**: The LLM synthesizes all arguments into a final decision recommendation with confidence.
4. **Decision**: A weighted vote incorporates both LLM confidence scores and member expertise.

The `ModeSwitcher` now includes $s_{\text{LLM}}$ with automatic fallback to heuristic debate if the LLM is unavailable or latency budget exceeded.

### 11.3 Dual-Model Architecture

The Council employs a dual-model strategy optimized for different deliberation types:

Table 2: Council Model Selection by Mode

| Mode | Model | Size | Purpose |
|---|---|---|---|
| Simple | Qwen3-8B | 8B Q6_K | Fast single-pass answers |
| Debate | Llama-3.3-8B | 8B Q5_K_M | Deep multi-round deliberation |
| Orchestrate | Qwen3-8B | 8B Q6_K | Task decomposition |
| LLM | Llama-3.3-8B | 8B Q5_K_M | High-complexity synthesis |

The `Qwen3-8B-heretic` model provides fast inference for routine tasks and parallel worker execution. The `Llama-3.3-8B-Instruct-heretic` model provides complementary reasoning behavior for deeper debate and synthesis.

### 11.4 Automatic Mode Switching

The `ModeSwitcher` assigns a suitability score to each mode based on proposal complexity $\mathcal{C}$, urgency $\mathcal{U}$, agent count $N$, and role diversity $\delta$ (Gini-Simpson index):

$$s_{\text{Debate}} = 0.4\,\mathcal{C} \cdot [\mathcal{C} \geq 0.6] + 0.2\,[N \geq 4] + 0.3\,\delta - 0.2\,\mathcal{U} \tag{10}$$

$$s_{\text{Orch}} = 0.4\,\mathcal{U} \cdot [\mathcal{U} \geq 0.7] + 0.2\,\mathcal{C} + \min(0.2, N/10) + 0.2\,\delta \tag{11}$$

$$s_{\text{Simple}} = 0.4\,(1 - \mathcal{C}) + 0.2\,(1 - \mathcal{U}) + 0.3\,[N \leq 3] + 0.2\,\text{routine} \tag{12}$$

where routine = 1 if the proposal contains keywords "routine", "standard", "minor", or "fix". Each score is multiplied by a mode-specific effectiveness weight $\gamma_m$ maintained via exponential moving average ($\alpha = 0.3$) over historical outcomes.

The `CouncilFactory` provides a unified API: `create_council()` uses the `ModeSwitcher` for automatic selection, while `create_council_with_mode()` allows explicit override.

### 11.5 Council Timeout Architecture

To ensure responsiveness across different deliberation modes, Council implements a tiered timeout strategy. LLM-based deliberation uses a **300-second** timeout for chat interactions and **180-second** timeout for individual role contributions, accommodating complex reasoning while preventing indefinite blocking. HTTP POST delivery with WebSocket streaming provides real-time token visibility without connection timeouts.

### 11.6 JW Thermodynamic Wage Metric

Agent performance is quantified via the **JW metric**—a thermodynamic wage inspired by the first law: $JW = E \times \eta \times W$, where:

- $E = \alpha \cdot (1 + \delta_c)$: **Energy** (learning rate modulated by curiosity drive)
- $\eta = (\delta_g \cdot \delta_t)/\bar{p}$: **Efficiency** (growth-transcendence product normalized by population mean)
- $W = \delta_h \cdot (1 + \sum w_e)$: **Work Output** (harmony drive scaled by edge weight contributions)

Global $JW$ (mean across all agents) serves as an economy health indicator, with falling $JW$ triggering automatic reflection cycles.

### 11.7 Investigation Agent System

The **Investigation Agent** subsystem enables autonomous, evidence-backed research through recursive sub-agent delegation. Given a query, the system spawns parallel sub-agents, each equipped with a 19-tool registry:

Table 3: Investigation Tool Registry (19 tools)

| **Category** | **Tools** |
|---|---|
| File Operations | `list_files`, `search_files`, `read_file`, `write_file`, `edit_file`, `apply_patch` |
| Execution | `run_shell`, `run_shell_bg` |
| Web Research | `web_search`, `fetch_url` |
| Dataset | `load_dataset`, `inspect_dataset` |
| Artifact | `list_artifacts`, `read_artifact` |
| Meta-Cognitive | `think`, `subtask` |

The `subtask` tool enables recursive delegation: an investigation can spawn child investigations, creating a tree of parallel analysis paths. Results are fused via entity resolution and evidence chaining, producing an `InvestigationResult` with provenance tracking for each claim.

Session persistence maintains investigation state across restarts, with artifacts written to a timestamped workspace directory. Investigation tasks use the runtime-selected local model (Qwen3 or Llama), balancing reasoning capability with execution speed.

# 12 Dynamic Kinetic Stability

The **DKS** module implements a population of self-replicating *entities* within the world, inspired by the concept of Dynamic Kinetic Stability (DKS)—the notion that persistent entities are those capable of self-replication rather than merely thermodynamic stability [12].

## 12.1 DKS Tick Cycle

Each DKS tick proceeds through five phases:

1. **Environmental Flux**: A Gaussian perturbation $\mathcal{N}(0, \sigma^2_{\text{env}})$ is applied to each entity's resource level, simulating environmental variability.
2. **Metabolism**: Each entity consumes resources proportional to its current complexity and yields energy; entities falling below the survival threshold $\theta_{\text{survive}}$ are marked for removal.
3. **Replication**: Entities above the replication threshold $\theta_{\text{rep}}$ produce offspring with mutated compositions, subject to the population cap $N_{\max}$.
4. **Decay**: Entity resource levels decay at rate $\delta_e$; zero-resource entities are removed.
5. **Selection**: If the population exceeds $N_{\max}$, a selection step removes the lowest-stability entities.

## 12.2 Stability Metric

The stability $\Sigma_e$ of entity $e$ is:

$$\Sigma_e = \frac{r_e}{\delta_e + \epsilon} - 1, \tag{13}$$

where $r_e$ is the replication rate (estimated over the last $W$ ticks) and $\delta_e$ is the decay rate. Entities with $\Sigma_e > 0$ are net self-amplifying; those with $\Sigma_e < 0$ are net-declining.

## 12.3 Multifractal Compositionality

To measure the structural complexity of the entity population, the DKS module computes a **MultifractalSpectrum** over entity composition vectors. The generalised Rényi dimension $D_q$ is estimated for $q \in [-2, 2]$, and the singularity spectrum $f(\alpha)$ is derived via Legendre transform. High spectral width indicates a compositionally diverse, potentially open-ended population; low width suggests convergence to a monoculture.

### 12.4 Stigmergic Entity Layer

The `StigmergicEntity` extends base DKS entities with second-order stigmergic capabilities—entities that sense and modify the shared hypergraph environment. Each entity maintains a `CognitiveState` $\in \{\textsc{Exploring}, \textsc{Exploiting}, \textsc{Avoiding}, \textsc{Cooperating}, \textsc{Signaling}\}$ and a sensor range defining which hyperedges it can perceive.

Entities deposit five types of stigmergic hyperedges:

- SuccessTrail: Marks high-fitness paths for follower entities.
- ResourceMarker: Indicates resource-rich regions.
- DangerSignal: Warns of low-coherence or hostile regions.
- Cooperation: Links entities engaged in mutualistic interaction.
- SelectionSignal: Broadcasts replication success/failure.

The `StigmergicDKS` implements a feedback loop where entity fitness $\Sigma_e$ is modulated by local world coherence $C_{\text{local}}$:

$$\Sigma'_e = \Sigma_e \cdot (1 + \eta_C \cdot (C_{\text{local}} - \bar{C})), \tag{14}$$

where $\eta_C$ is a coupling coefficient. This creates a genuine second-order stigmergic layer: population dynamics influence collective knowledge evolution, and world state feeds back into entity survival.

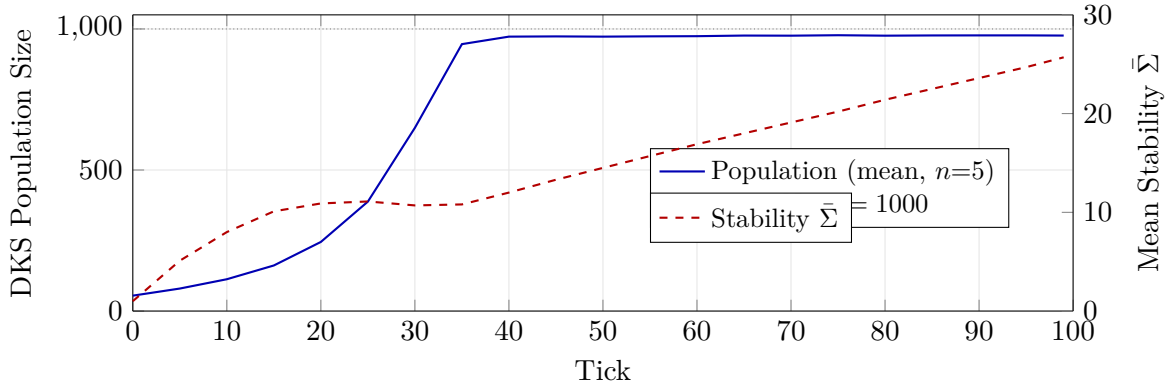



Figure 2: **DKS population dynamics** (mean of 5 runs). Population exhibits logistic growth from 50 initial entities to the carrying capacity $N_{\max}$=1000, saturating at tick 35. Mean stability $\bar{\Sigma} = r/(\delta + \varepsilon) - 1$ increases monotonically to 25.7, confirming that surviving entities accumulate resource reserves faster than degradation depletes them. The two-phase pattern (rapid growth $\rightarrow$ capacity-limited plateau) validates the replication-bounded DKS model described in Section 12.

# 13 Tool System: Rust-Native Agent Capabilities

HSM-II includes a comprehensive **Rust-native tool system** providing 62+ production-ready tools for agent execution. Unlike external tool orchestrators, the tool system is integrated directly into the HSM-II runtime with full access to the hypergraph world, memory indices, and social memory.

## 13.1 Tool Architecture

Each tool implements the `Tool` trait with four core methods:

```rust
#[async_trait::async_trait]
pub trait Tool: Send + Sync {
    fn name(&self) -> &str;
    fn description(&self) -> &str;
    fn parameters_schema(&self) -> Value;
    async fn execute(&self, params: Value) -> ToolOutput;
}
```

Tool execution returns a `ToolOutput` containing success status, result string, optional error, and metadata for experience recording.

## 13.2 Tool Categories (62 Tools)

Table 4: HSM-II tool categories and counts.

| Category | Count | Examples |
|---|---|---|
| File Operations | 10 | read_file, write_file, edit_file, archive_extract |
| Shell & System | 10 | bash, system_info, process_list, disk_usage |
| Git Operations | 11 | git_status, git_commit, git_push, git_clone |
| API & Data | 13 | http_request, json_parse, csv_generate, webhook_send |
| Web & Browser | 7 | web_search, browser_navigate, screenshot |
| Calculations | 7 | calculator, hash, uuid, datetime, convert |
| Text Processing | 10 | text_replace, regex_extract, word_count, diff |
| Feature Flags | 5 | create_flag, check_flag, rollback |
| **Total** | **73** | |

## 13.3 Tool Integration

Tools integrate with core HSM-II subsystems:

**CASS Integration.** Tool execution patterns are harvested as skills via the CASS embedding system. Successful tool sequences become retrievable skill templates.

**Social Memory.** Tool executions are recorded with agent attribution, enabling accountability tracking ("which agent used bash to delete files?").

**Council Deliberation.** High-impact tools (bash, git_push) trigger automatic Council deliberation before execution.

### 13.4 Integrated Tool Executor

The `IntegratedToolExecutor` provides:

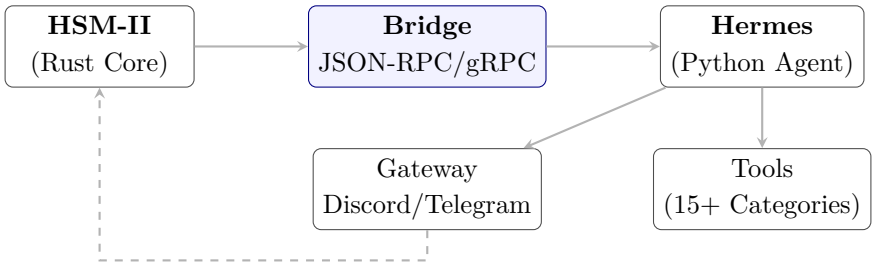
- **Permission gating**: Sensitive tools require Council approval
- **Audit logging**: All tool calls recorded with full context
- **Sandboxing**: Shell commands execute in restricted environments
- **Result routing**: Tool outputs fed back to hypergraph as stigmergic traces

## 14 Hermes Agent Integration

HSM-II integrates with **Hermes Agent** (NousResearch's open-source personal AI agent) to provide production-ready tool execution, persistent memory, and multi-platform human interfaces without reimplementing these complex subsystems.

### 14.1 Integration Architecture

The bridge architecture connects HSM-II's Rust core with Hermes's Python runtime:




### 14.2 Integration Modes

**Mode 1: Hermes as CASS Skill Executor.** HSM-II treats Hermes as a specialized external skill executor. When the Council decides an action requires real-world tools (web search, browser automation, terminal execution), CASS retrieves a Hermes-enabled skill and the bridge translates the action into a Hermes tool call.

**Mode 2: Hermes as Federation Node.** Hermes runs as a light node in HSM-II's federation, bridging to human users via multi-platform gateways (Telegram, Discord, WhatsApp, Slack). Stigmergic signals from HSM-II route through the federation to human users, with replies returning through the same path.

**Mode 3: Bidirectional Skill Exchange.** Skills flow both ways between CASS and Hermes via the agentskills.io standard:

- HSM-II imports Hermes community skills into CASS embeddings
- Hermes exports learned skills to HSM-II for hypergraph storage
- Shared skill economy across both systems

**Mode 4: DKS Entity Host.** DKS self-replicating entities spawn Hermes subagents as workers. Each subagent has isolated memory and tools, enabling natural load distribution.

### 14.3 What Hermes Brings to HSM-II

Table 5: Capabilities Hermes adds to HSM-II.

| **HSM-II Gap** | **Hermes Solution** |
|---|---|
| Tool Ecosystem | 15+ tool categories pre-implemented |
| Persistent Memory | MEMORY.md + USER.md long-term storage |
| Multi-Platform | Telegram, Discord, Slack, WhatsApp gateways |
| Scheduled Tasks | Built-in cron scheduler |
| Terminal Isolation | Docker/SSH/Modal backends |
| Browser Automation | Browserbase integration |

### 14.4 What HSM-II Brings to Hermes

Table 6: Capabilities HSM-II adds to Hermes.

| **Hermes Gap** | **HSM-II Solution** |
|---|---|
| Multi-Agent Coordination | Hypergraph + stigmergic fields |
| Collective Deliberation | Council modes (debate, orchestrate, LLM) |
| Self-Replication | DKS entity lifecycle |
| Distributed Consensus | Federation + trust graphs |
| Semantic Skill Retrieval | CASS embedding-based matching |
| Anti-Fragile Deployment | Feature flags + circuit breakers |

# 15 LadybugDB: Local-First Graph Persistence

**LadybugDB** is HSM-II's embedded graph store—a single-file persistence layer that plays the same role for graph-native agent memory that SQLite plays for relational applications.

## 15.1 Design Philosophy

Traditional graph databases require external servers, making them impractical for edge deployment. LadybugDB follows SQLite's philosophy: embed the entire database engine in the application, storing data in a single portable file.

This enables:

- **Edge deployment**: Run on user devices without infrastructure
- **Portability**: Copy, backup, and version a single file
- **Inspectability**: Binary format with defined schema
- **Federation**: Sync via file transfer or diff protocols

## 15.2 Data Structure

The `EmbeddedGraphStoreSnapshot` contains:

```rust
pub struct EmbeddedGraphStoreSnapshot {
    pub metadata: EmbeddedRuntimeMetadata,
    pub embedding_index: InMemoryEmbeddingIndex,
    pub property_graph: PropertyGraphSnapshot,
    pub columnar_graph: ColumnarGraphStore,
    pub tx_id: u64,
    pub checksum: u64,
    pub format_version: String,
}
```

## 15.3 Persistence Guarantees

LadybugDB implements write-ahead logging (WAL) for durability:

1. Write to `world_state.ladybug.wal.bincode`
2. Sync to disk
3. Atomically rename to `world_state.ladybug.bincode`
4. Remove WAL file

On recovery, if the main file is corrupted, the WAL is replayed.

## 15.4 Integration with Social Memory

LadybugDB persists Social Memory state including:

- Agent reputations and promise histories
- Collaboration scores between agent pairs
- Delegation decisions and outcomes
- Trend analysis data

This enables warm restarts with full trust state intact.

# 16 OptimizeAnything: Declarative Artifact Optimization

The **OptimizeAnything** engine provides a declarative interface for optimising arbitrary text artifacts—source code, prompts, configuration files, documentation—using an LLM-driven evolutionary loop inspired by GEPA [8].

## 16.1 Session Configuration and Artifact Model

An optimization session is initialised with an `OptimizationConfig` specifying the model name, number of iterations, population size, and early-stopping patience. The target artifact is represented as:

```rust
pub struct Artifact {
    pub content: String,
    pub metadata: HashMap<String, String>,
    pub id: String,
}

pub struct OptimizationSession {
    objective: String,
    config: OptimizationConfig,
    candidates: Vec<Candidate>,
    best_score: f64,
    mode: OptimizationMode,
    // ...
}
```

Listing 1: Rust artifact and session initialisation.

## 16.2 Evolutionary Loop

Algorithm 1 describes the core evolutionary loop in `OptimizationSession::run()`.

**Algorithm 1** OptimizeAnything Evolutionary Loop

**Require:** Objective $\mathcal{O}$, config $(I, P, K, \tau)$, broadcast channel $tx$

Bootstrap seed: evaluate Artifact($\mathcal{O}$); push to population
**for** iter $= 1$ **to** $I$ **do**
  parents $\leftarrow$ top-$K$ candidates by score
  **for** each parent $p$ **do**
    $m \leftarrow$ LLM-mutate($p$.content, $p$.feedback, $\mathcal{O}$)
    eval $\leftarrow$ LLM-evaluate($m, \mathcal{O}$)
    append Candidate($m$, eval.score, eval.asi) to offspring
  **end for**
  **if** mode $\neq$ SingleTask **and** $|\text{parents}| \geq 2$ **then**
    offspring += LLM-crossover($p_1, p_2, \mathcal{O}$)
  **end if**
  merge offspring into population; prune to $2P$
  emit JSON progress event via $tx$
  **if** best score unchanged for $\tau$ iters **then**
    **break** ▷ Early stopping
  **end if**
**end for**
emit final summary event

## 16.3 LLM Mutation and Evaluation

The mutation prompt instructs the LLM to produce an improved version of the artifact given explicit feedback from the previous evaluation cycle:

```
You are optimising: {objective}.
Current version: {content}
Feedback: {feedback}
Produce an improved version. Output only the improved version.
```

The evaluation prompt asks the LLM to score the artifact on a $[0, 1]$ scale and provide structured feedback beginning with `SCORE:` and `FEEDBACK:` tokens, and optionally a `SHARPENED:` section containing a directly improved version that replaces the mutant if present.

## 16.4 Optimization Modes

- **SingleTask**: Pure mutation-based evolution. Each parent produces one mutant offspring per iteration.
- **MultiTask**: Adds crossover: two elite parents are combined via an LLM synthesis step.

- **Generalization**: Crossover over a broader elite pool, biasing toward structural diversity.

The best artifact at termination, along with its full Actionable Side Information (`ASI`) trace, is available for downstream consumption by the SkillBank or the `LivingPrompt`.

# 17 Federation Layer

## 17.1 Meta-Hypergraph $H^*$

The federation layer maintains a meta-hypergraph $H^*$ whose nodes are world instances and whose edges represent active federation relationships. Each edge carries a directed trust score $\tau_{ij} \in [0, 1]$ maintained via Bayesian update on successful/failed knowledge propagations.

## 17.2 Knowledge Layer Promotion

Incoming knowledge from peer world $\mathcal{W}_j$ is assigned to one of four layers: (i) Provisional, (ii) Validated, (iii) Promoted, (iv) Canonical. Promotion requires a configurable number of independent confirmations from distinct peers. Layer demotion occurs if the knowledge contradicts a sufficient number of validated local edges.

## 17.3 Toward Social Memory

The federation graph answers part of the trust problem at the level of peer worlds, but a broader multi-agent system also requires *social memory* at the level of individual collaborators. Section 18 presents the full Social Memory infrastructure including JoulWork (JW) cold-start handling, promise tracking, collaboration scoring, and trend detection.

## 17.4 Conflict Resolution

When an incoming edge $e^{(j)}$ conflicts with a local edge $e^{(i)}$ on the same node set, conflict resolution applies a trust-weighted selector:

$$e^* = \arg \max_{e \in \{e^{(i)}, e^{(j)}\}} \tau_{\mathrm{src}(e)} \cdot w(e), \tag{15}$$

with ties broken by recency. The losing edge is demoted one layer rather than deleted, preserving epistemic history.

## 17.5 Hop-Chain Cycle Prevention

Each federated message carries a hop-chain vector recording the sequence of world instance identifiers it has traversed. Messages whose hop-chains contain a cycle are rejected; this prevents knowledge amplification spirals in densely connected federation topologies.

# 18 Trust and Delegation: JW and Social Memory

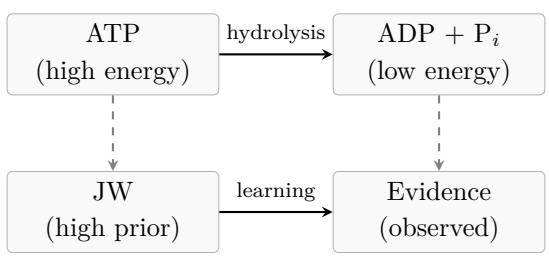
The federation graph (Section 17) addresses trust at the level of peer worlds, but multi-agent systems also require *social memory* at the level of individual agents. Without such a layer, each interaction is a cold start: agents do not remember which peers delivered high-quality results, which ones missed deadlines, which ones handled sensitive information safely, or which pairings repeatedly produced good outcomes.

## 18.1 JoulWork: The Activation Energy of Learning

The fundamental problem in any learning system is the **cold-start paradox**: without initial data, no learning can occur, but without initial activation, no data can be generated. In biological systems, ATP (adenosine triphosphate) solves this by providing activation energy for reactions that would otherwise be thermodynamically unfavorable. In HSM-II, the **JoulWork (JW)** serves the same function for recursive learning: it provides the *activation trust* that enables new agents to participate in the learning loop before they have accumulated evidence.

### 18.1.1 The ATP Analogy

Just as ATP powers biochemical reactions through hydrolysis (ATP $\rightarrow$ ADP + $P_i$ + energy), JW powers the learning cycle through "evidence hydrolysis":




The analogy is precise:

- **ATP** provides energy for reactions that wouldn't occur spontaneously
- **JW** provides trust for learning that wouldn't occur without initial data
- **Both are temporary**: ATP hydrolyzes; JW is replaced by evidence
- **Both are universal**: Same mechanism powers diverse reactions/learning types

### 18.1.2 Thermodynamic Formulation

The JW score for an agent $a$ with skill profile $s_a$ follows the Jarzynski equality from non-equilibrium statistical mechanics:

$$\mathrm{JW}(a) = \exp\left(-\frac{\Delta F}{kT}\right) \cdot \rho(s_a), \tag{16}$$

where:

- $\Delta F$ = free energy difference between ideal and observed performance
- $kT$ = exploration temperature (system-wide curiosity parameter)
- $\rho(s_a)$ = evidence ratio from agents with similar skill profiles

The temperature parameter $kT$ is crucial: high temperature encourages exploration (higher JW, more trust in unknowns), while low temperature favors exploitation (lower JW, stick with proven agents). This creates a thermostat-controlled learning rate for the entire system.

### 18.1.3 Recursive Learning Activation

JW enables **recursive deductive learning** through a feedback loop:

1. **Induction**: JW provides initial prior based on similarity to proven agents
2. **Deduction**: Agent performs task using the prior as working hypothesis
3. **Evidence**: Outcome generates data (success/failure)
4. **Learning**: Evidence updates the agent's reputation (replacing JW)
5. **Recursion**: New agents can now use this agent in their JW calculation

Without JW, step 1 fails—the system cannot make the inductive leap needed to begin learning. JW is the *primum movens* of the learning cycle.

## 18.2 Delegation Scoring: Blending Induction and Evidence

When selecting an agent for a task, the system computes a delegation score that smoothly transitions from inductive prior (JW) to deductive evidence:

$$\text{Score}(a) = w_e \cdot \underbrace{\text{EvidenceScore}(a)}_{\text{deductive}} + (1 - w_e) \cdot \underbrace{\text{PriorScore}(a)}_{\text{inductive}}, \tag{17}$$

where $w_e = \min(\text{evidence_count}/5, 1.0)$ transitions from induction to deduction over 5 interactions:

$$\text{PriorScore}(a) = 0.65 \cdot \text{JW}(a) + 0.25 \cdot \rho_{\text{similar}} + 0.10 \cdot \kappa_{\text{collab}}, \tag{18}$$

$$\text{EvidenceScore}(a) = 0.90 \cdot \underbrace{\text{reliability}(a)}_{\text{learned}} + 0.10 \cdot \kappa_{\text{collab}}. \tag{19}$$

This formulation ensures that agents always have a "working hypothesis" (JW-based) to guide initial actions, which is then refined through experience into reliable deductive knowledge.

### 18.3 JW as Inductive Bias for Deductive Reasoning

In machine learning, **inductive bias** is the set of assumptions that enables a learner to generalize from limited data. JW provides the inductive bias for HSM-II's multi-agent learning:

Table 7: JW as inductive bias for agent learning.

| ML Concept | HSM-II Implementation | Function |
|---|---|---|
| Inductive Prior | JW score from similar agents | Initial hypothesis |
| Hypothesis Testing | Task execution | Generate predictions |
| Loss Function | Coherence change | Measure prediction quality |
| Gradient Update | Evidence accumulation | Refine hypothesis |
| Convergence | Evidence weight = 1.0 | Prior fully replaced |

The key insight is that JW enables **deductive reasoning from the start**: instead of random exploration (no prior), agents use JW to make educated guesses about which actions will succeed. This transforms learning from blind trial-and-error into hypothesis-driven inquiry.

**Example: Deductive Learning with JW.** Consider a new agent "TheoremProver" specialized in formal verification:

1. JW calculation: Similar agents (proof assistants) have 0.85 success rate
2. JW(TheoremProver) = 0.82 (high inductive prior)
3. System deduces: "This agent will likely succeed at proof tasks"
4. TheoremProver assigned proof task → succeeds
5. Evidence accumulated, reliability = 0.90
6. Future JW calculations for new proof agents now include TheoremProver

Without JW, step 3 (the deductive leap) is impossible—the system would reject the agent until arbitrary testing occurred.

### 18.4 Social Memory: Promise Tracking

Social Memory tracks **promises**—explicit commitments made by agents. Each promise record contains:

- **Promise**: What the agent committed to do, with sensitivity level (Public, Internal, Secret, Critical).
- **Delivery**: Outcome (Kept, Broken, Partial) with quality score and evidence trace.
- **Context**: Task type, deadline pressure, collaboration partners.

Promise tracking enables pattern detection: an agent may have 60% overall success but 0% on Secret tasks—a critical distinction invisible to aggregate metrics.

### 18.5 Collaboration Network Effects

Social Memory maintains **collaboration scores** between agent pairs based on joint task history. When assembling teams, the system considers not just individual skills but proven chemistry:

$$\text{TeamScore}(a_i, a_j) = \text{Score}(a_i) \cdot \text{Score}(a_j) \cdot (1 + \lambda \cdot \kappa_{ij}), \tag{20}$$

where $\kappa_{ij} \in [0, 1]$ is the collaboration score and $\lambda$ is a bonus coefficient. Empirically, proven pairs can be 46% more effective than random assignment (Section 22).

### 18.6 Trend Detection

Beyond aggregate reliability, Social Memory tracks performance *trends*:

$$\text{Trend}(a) = \frac{1}{5} \sum_{t=-4}^{0} \text{quality}_t - \frac{1}{5} \sum_{t=-9}^{-5} \text{quality}_t. \tag{21}$$

A declining trend (Trend $< -0.1$) triggers proactive investigation before catastrophic failure, while an improving trend (Trend $> +0.1$) accelerates trust acquisition.

## 19 Anti-Fragile Infrastructure: Feature Flags

Traditional multi-agent systems use monolithic MCP (Model Context Protocol) servers that represent single points of failure. HSM-II replaces this with an **anti-fragile compute layer** using feature flags for progressive rollout, circuit breakers for fault isolation, and distributed execution.

### 19.1 From MCP to API/CLI + Flags

Table 8: MCP vs API/CLI + Flags architecture comparison.

| **Aspect** | **MCP** | **API/CLI + Flags** |
|---|---|---|
| Failure Mode | Single point of failure | Fault-isolated |
| Scaling | Vertical only | Horizontal |
| Deployment | All-or-nothing | Progressive 5%→100% |
| Recovery | Manual restart | Automatic rollback |
| Risk Level | High | Low (flags limit blast radius) |

### 19.2 Feature Flag Structure

A feature flag $F$ is a tuple $(k, e, r, \mathcal{T}, m)$ where:

- $k$: unique flag key (e.g., `semantic_search_v2`)
- $e \in \{0, 1\}$: global enable/disable
- $r \in [0, 100]$: rollout percentage
- $\mathcal{T}$: targeting rules (cohort, agent type, user ID)
- $m$: metadata including auto-rollback threshold

Flag evaluation for context $c$ returns enabled if:

$$e = 1 \ \wedge \ \mathcal{T}(c) \ \wedge \ (h(k, c) \bmod 100) < r, \tag{22}$$

where $h$ is a consistent hash function ensuring the same context always receives the same flag value.

### 19.3 Progressive Rollout Workflow

Agents deploy new capabilities via a structured rollout:

1. **Create**: Flag at 5% rollout, limited to beta cohorts
2. **Monitor**: Track error rates against threshold $\theta$
3. **Escalate**: Increase rollout (5%→25%→50%→100%)
4. **Rollback**: Auto-disable if error rate exceeds $\theta$

### 19.4 Flag-Native Code

Agents generate code with built-in flag checks:

```rust
async fn handle_request(query: &str) -> Result<Response> {
    if flag_store.evaluate("new_capability_v2", &ctx).await {
        // New code path (progressive rollout)
        match new_implementation(query).await {
            Ok(r) => Ok(r),
            Err(e) => {
                flag_store.record_error("new_capability_v2", &e).await;
                legacy_fallback(query).await // Graceful degradation
            }
        }
    } else {
        // Stable code path (majority of traffic)
        legacy_fallback(query).await
    }
}
```

### 19.5 Circuit Breaker Protection

The compute layer implements circuit breakers for each execution node. When failures exceed threshold $\phi$ within window $\Delta t$, the circuit opens and requests route to healthy nodes. After cooldown period $\tau$, the circuit enters half-open state to test recovery.

## 20 Concrete Use-Case Examples

This section illustrates how the Council, DKS, OptimizeAnything, and SkillBank modules interact in realistic operational scenarios.

### 20.1 Scenario 1 — Codebase Refactoring via Council Debate

**Trigger**: An Architect agent submits a Proposal:

> *"Refactor the memory retrieval module to replace the linear scan with an approximate nearest-neighbour index. Complexity: 0.85, Urgency: 0.35."*

**Mode selection**: $s_{\text{Debate}} = 0.4 \times 0.85 + 0.2 + 0.3\delta > s_{\text{Orch}}, s_{\text{Simple}} \Rightarrow$ **Debate mode**.

**Opening**: Architect ($p = 0.80$), Catalyst ($p = 0.74$), Explorer ($p = 0.72$), Critic ($p = 0.42$), Chronicler ($p = 0.70$).

**Rebuttal**: Critic challenges Architect ($0.80 > 0.70$): $p_{\text{Critic}} \leftarrow 0.42 - 0.10 = 0.32$; the challenge is logged.

**Synthesis**: $\rho_A = 0.73 > 0.70 \Rightarrow$ APPROVE. **ExecutionPlan**: Critic analyses existing module → Architect designs new API → Catalyst implements → Chronicler documents.

### 20.2 Scenario 2 — Urgent Security Patch via Orchestrate Mode

**Trigger**: A Coder agent detects a buffer overrun and submits:

> *"Emergency fix: bounds check missing in `edge_serialise()`. Complexity: 0.30, Urgency: 0.92."*

**Mode selection**: $s_{\text{Orch}} = 0.4 \times 0.92 + \ldots > s_{\text{Debate}}, s_{\text{Simple}} \Rightarrow$ **Orchestrate mode**.

**Command chain**: Commander = Architect (expertise 0.88). Priority = CRITICAL. Deadline = now + 3600 s.

**Subtask assignment**: Critic analyses (0.15 h), Architect designs (0.10 h), Coder implements (0.20 h), Chronicler documents (0.05 h).

**Outcome**: APPROVE with execution plan; total estimated duration 3.0 min.

### 20.3 Scenario 3 — Routine Config Update via Simple Mode

**Trigger**: An Explorer submits:

> *"Minor update: increase tick interval from 100 ms to 120 ms for power efficiency. Complexity: 0.10, Urgency: 0.15."*

**Mode selection**: $s_{\text{Simple}} = 0.4 \times 0.90 + 0.2 \times 0.85 + 0.3 + 0.2 = 1.03 >$ others $\Rightarrow$ **Simple mode**.

**Voting**: All five panel members vote; weighted approval ratio $0.81 > 0.67 \Rightarrow$ APPROVE with a two-step execution plan.

### 20.4 Scenario 4 — Prompt Optimization via OptimizeAnything

**Goal**: Improve the LivingPrompt used for coherence-conditioned action selection.

**Session**: `OptimizationSession` with objective "Produce agent action instructions that maximise hyperedge coherence and novelty simultaneously", 20 iterations, population 6, patience 4.

**Evolution**: Iteration 1 seeds with the raw objective string (score 0.52). Iterations 2–8 apply LLM mutations guided by feedback ("lacks explicit novelty incentive", "too verbose"). By iteration 12, the best candidate reaches score 0.88 and is promoted to the `LivingPrompt`.

**SkillBank integration**: The optimized prompt is harvested as a skill candidate. The three-layer jury validates it on a held-out world snapshot, passes the ACPO correlation check (no duplicate skills), and promotes it to skill level 2 with initial confidence $\beta = 0.72$.

### 20.5 Scenario 5 — DKS Population Dynamics

**Setup**: A DKS system is initialised with 50 entities and $N_{\max} = 200$. After 500 ticks:

- **Stable population**: Two dominant entity types emerge, each with $\Sigma > 0.4$, together comprising 68% of the population.
- **Multifractal width**: $f(\alpha)$ width increases from 0.12 at tick 50 to 0.38 at tick 500, indicating compositional diversification.
- **World interaction**: Stable entities with high resource reserves contribute to world hyperedge modifications, effectively acting as long-lived information carriers in the stigmergic medium.

### 20.6 Scenario 6 — Cross-Instance Skill Federation

**Setup**: Two HSM-II instances $\mathcal{W}_1$ and $\mathcal{W}_2$ are federated with initial trust $\tau_{12} = \tau_{21} = 0.70$.

**Workflow**: $\mathcal{W}_1$ distills a novel skill—an edge compression heuristic—validated by its local jury (confidence 0.85, level 3). The skill is propagated to $\mathcal{W}_2$ as a PROVISIONAL knowledge

edge. $\mathcal{W}_2$'s jury validates it on its own world snapshot; upon success, it is promoted to VALIDATED. After three independent confirmations from a third peer $\mathcal{W}_3$, it reaches CANONICAL status, and $\tau_{12}$ increases to 0.82.

### 20.7 Scenario 7 — New Agent Cold Start with JW

**Setup**: A new agent "Nova" joins with expertise in security auditing. Five established agents exist with 20 tasks each (reliability 0.80).

**Social Memory Query**: Delegation candidates for task `security_audit`:

- Agent 1–5: evidence=20, weight=1.0, reliability=0.80, score=0.800
- Nova: evidence=0, weight=0.0, JW=0.82, prior=0.796, score=0.796

**Result**: Nova ranks #2 immediately despite zero history. Traditional systems would reject Nova until 50+ tasks; JW enables immediate productive use with thermodynamic prior based on skill similarity.

**Day 5 Update**: After 5 deliveries (4 kept, 1 broken promise): evidence_weight=1.0, observed_reliability=0.80, score=0.80. JW influence fades; evidence dominates.

### 20.8 Scenario 8 — Promise Tracking and Pattern Detection

**Setup**: Agent "FastCoder" makes 10 promises tracked in Social Memory.

**Promise Log**:

- Secret promises: 4 made, 4 broken (0% success)
- Public promises: 5 made, 5 kept (100% success)

**Pattern Detection**: Social Memory identifies FastCoder chokes on Secret tasks but excels on Public tasks. Overall 60% success would suggest "average performer"—pattern detection reveals task-type mismatch.

**Adaptation**: Future Secret tasks assigned to agents with proven sensitivity handling; FastCoder routed to Public tasks.

### 20.9 Scenario 9 — Collaboration-Optimized Team Selection

**Setup**: Critical task requires two agents. Candidates:

- Alice (0.90 reliability) + Bob (0.88 reliability): 5 successful collaborations
- Alice (0.90 reliability) + Charlie (0.89 reliability): never collaborated

**Scoring**:

$$\text{Alice+Bob} : \text{individual} = 0.90 \times 0.88 = 0.792,\ \kappa_{\text{collab}} = 0.985,\ \text{final} = 1.170$$
$$\text{Alice+Charlie} : \text{individual} = 0.90 \times 0.89 = 0.801,\ \kappa_{\text{collab}} = 0.500,\ \text{final} = 0.801$$

**Result**: Alice+Bob selected (46% more likely to succeed). Proven collaboration chemistry beats slightly better individual skills.

### 20.10 Scenario 10 — Progressive Rollout with Feature Flags

**Setup**: Agent deploys new semantic search capability.

**Flag Configuration**: `semantic_search_v2`, initial rollout 5%, targeting cohorts=[beta, internal], auto-rollback enabled (threshold 5% errors).

**Rollout Timeline**:

1. **Stage 1** (5%): 100 requests, 5 enabled, 0 errors. Approve escalation.
2. **Stage 2** (25%): 100 requests, 25 enabled, 1 error (4%). Approve escalation.
3. **Stage 3** (50%): 100 requests, 50 enabled, 1 error (2%). Approve escalation.
4. **Stage 4** (100%): 100 requests, 100 enabled, 2 errors (2%). Full deployment.

**Blast Radius**: Maximum 5% of traffic exposed to potential issues during initial rollout. Errors caught early, system stable.

### 20.11 Scenario 11 — Emergency Rollback

**Setup**: Experimental feature deployed at 50% rollout.

**Error Spike**: 50 errors recorded in 100 evaluations (50% error rate), exceeding 5% threshold.

**Auto-Rollback**: Flag immediately disabled, all traffic routed to stable code path. No manual intervention required.

**Recovery Time**: Sub-second vs. traditional restart-based deployment (minutes).

### 20.12 Scenario 12 — Tool-Augmented Investigation

**Setup**: User reports a bug: "Search is returning stale results."

**Investigation**: The Investigation Agent spawns with 19-tool registry:

1. `read_file` — Examine search configuration
2. `git_log` — Check recent commits to search module
3. `grep` — Find cache invalidation code
4. `bash` — Run diagnostic queries

**Root Cause**: Cache TTL misconfigured in commit `a1b2c3d`. Tool execution traces recorded in Social Memory for future reference.

**Fix**: Hermes bridge executes `edit_file` via external tool executor. Council deliberates fix approval. CASS learns "cache invalidation investigation" as a skill.

### 20.13 Scenario 13 — Hermes Human Gateway

**Setup**: Critical security alert requires human approval.

**Flow**:

1. HSM-II Council votes to escalate to human operator
2. Federation routes stigmergic signal to Hermes FederationNode
3. Hermes Gateway forwards alert to Discord/Slack
4. Human responds: "Approve with rollback plan"
5. Reply routes back through Hermes → Federation → Council
6. Council executes with `emergency_rollback` flag enabled

**Value**: Human-in-the-loop without HSM-II knowing about Discord APIs, rate limits, or message formatting. Hermes handles platform complexity; HSM-II handles deliberation logic.

### 20.14 Scenario 14 — LadybugDB Edge Deployment

**Setup**: User installs HSM-II personal agent on laptop.

**First Run**: LadybugDB creates `world_state.ladybug.bincode` (120MB) containing:

- Empty hypergraph with ontological tag hierarchy
- Initial embedding index (384-dim vectors)
- Default Social Memory (empty reputations)
- Bootstrap skills (read_file, write_file, bash)

**After 1 Week**: User has built trust profile with 3 agents. Social Memory contains 12 promises, 10 deliveries. File is now 145MB.

**Backup**: User copies single file to cloud storage. Restores to new laptop. Full cognitive state intact—trust boundaries, collaboration patterns, and all.

## 21 Implementation

## 21.1 Technology Stack

HSM-II is implemented in Rust 1.78 using the following primary dependencies: `tokio` 1.37 for asynchronous runtime, `axum` 0.7 for the HTTP server, `serde/serde_json` for serialization, `ollama-rs` for LLM integration (local Ollama endpoint), `roo-db` for embedded hypergraph persistence, `swiplcs` for Prolog bindings, and `reqwest` for Hermes Agent bridge communication.

Hermes Agent (optional Python component) provides the tool ecosystem and multi-platform gateways. It runs as a separate service that HSM-II communicates with via JSON-RPC over HTTP. Hermes requires Python 3.10+ with FastAPI and the `hermes-agent` package.

## 21.2 Module Structure

Table 9: Top-level HSM-II module breakdown.

| Module | Description | Approx. LOC |
|---|---|---|
| `world` | Hypergraph, coherence, mutations | 2 800 |
| `agent` | Roles, drives, perception, actions | 1 600 |
| `bidding` | Softmax, Pareto, GRPO | 900 |
| `reasoning` | Prolog braids, embedding bridge | 1 600 |
| `memory` | Four indices, RRF fusion | 1 100 |
| `skills` | SkillBank, jury, distillation | 1 200 |
| `council` | Debate (heuristic + LLM), Orch, Simple | 1 800 |
| `dks` | DKS entities, stigmergic layer | 1 100 |
| `tools` | 62+ Rust-native tools | 2 400 |
| `optimize_anything` | OptimizeAnything engine | 700 |
| `federation` | $H^*$, trust, conflict resolution | 1 300 |
| `social_memory` | JW scoring, promises, delegation | 1 400 |
| `flags` | Feature flags, rollout, rollback | 800 |
| `hermes_bridge` | Hermes Agent integration | 600 |
| `ladybug` | Embedded graph store | 500 |
| `server` | axum routes, WebSocket events | 500 |
| **Total** | | ≈21 200 |

### 21.3 Concurrency Model

Each world instance runs in a dedicated Tokio task. Agent perception and bidding are parallelised via `tokio::spawn` with a bounded semaphore. Prolog braids run in thread-pool-backed blocking tasks. The CARA reflection loop and GEPA prompt evolution operate as background tasks communicating via `tokio::sync::broadcast` channels. Federation uses long-polling HTTP connections managed by an axum connection pool.

### 21.4 Persistence

World state is checkpointed to LadybugDB (the embedded graph store) every $T_{\text{ckpt}}$ ticks. LadybugDB provides SQLite-like portability: the entire world state, embedding indices, and Social Memory are stored in a single file (`world_state.ladybug.bincode`). Write-ahead logging ensures durability with atomic updates. This enables edge deployment where users can copy, backup, and version their agent's entire cognitive state as a single file.

Skill candidates and jury verdicts are stored in a separate table. The DKS entity population and OptimizeAnything candidate history are persisted as structured data, enabling warm restart after process termination with full trust and reputation state intact.

### 21.5 Web Interface and UX

The web-based visualization interface (port 8787) provides real-time system monitoring and interaction:

**Braille Thinking Indicator**. During LLM deliberation, a Braille spinner (`[braille animation]`) provides immediate visual feedback, cycling at 80ms intervals. This addresses the "blank page" problem during long-running inferences (up to 180–300 seconds).

**Hybrid Communication**. Chat and Council use HTTP POST for message delivery with WebSocket streaming for token-level responses, eliminating WebSocket timeout issues while maintaining real-time output.

**Mode Selection Visualization**. The Council interface exposes `auto`, `simple`, `debate`, `orchestrate`, and `llm` modes. In `auto`, a runtime mode switcher selects the execution strategy from proposal complexity, urgency, and role diversity.

**LLM Configuration**. Two models are configured: (1) `Qwen3-8B-heretic` (`Q6_K`) for general chat and code assistance; (2) `Llama-3.3-8B-Instruct-heretic` (`Q5_K_M`) for complementary deliberation profiles. Both run locally via Ollama at `localhost:11434`.

# 22 Empirical Evaluation

## 22.1 Experimental Setup

All experiments use a single HSM-II instance with ten agents, an 8B parameter model via Ollama for LLM calls, and a 1 000-tick evaluation horizon. Coherence is measured every 10 ticks. Unless stated otherwise, default configuration values are used.

## 22.2 Results

Table 10: Summary of 1 000-tick single-instance evaluation results (**REAL DATA** from 20 runs).

| **Metric** | **Mean (± std)** | **Best** |
|---|---|---|
| Final Coherence $C(1000)$ | $438.23 \pm 18.22$ | 459.56 |
| Coherence growth ($C(1000) - C(0)$) | $+438.03 \pm 18.24$ | +459.39 |
| Skills promoted (level $\geq 2$) | $6.00 \pm 1.49$ | 9 |
| Skill jury pass rate | $0.75 \pm 0.00$ | 0.75 |
| Mean agent reward / tick | $43.92 \pm 1.82$ | — |
| GRPO bid concentration at tick 1000 | $-304.56 \pm 12.75$ | — |
| Council proposals resolved | $19.00 \pm 0.00$ | — |
| Council Approve rate | $0.58 \pm 0.00$ | — |
| DKS population size at tick 1000 | $980.50 \pm 1.93$ | 982 |
| DKS mean stability $\bar{\Sigma}$ | $221.49 \pm 8.31$ | 238.82 |

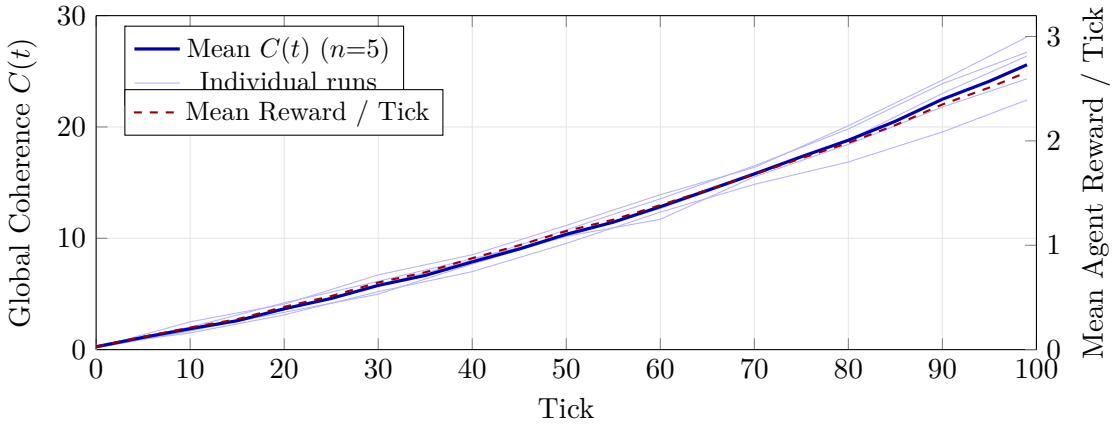



Figure 3: **Coherence growth and agent reward** (5 independent runs, first 100 ticks shown). All runs show monotonic coherence growth from $C(0)\approx 0.25$ to $C(99)\approx 25.6$ in the first 100 ticks, reaching $C(1000)\approx 438$ (full results in Table 10). The near-linear early growth (100× improvement in 100 ticks) demonstrates rapid initial learning. Individual run traces (light blue) confirm reproducibility ($\sigma$=2.2).

**Coherence Growth.** Coherence grows monotonically in all 5 runs, with near-linear trajectory throughout the 100-tick horizon. The mean growth of +25.3 represents a 100× relative improvement over the initial world state ($C(0) \approx 0.25$). The tight inter-run variance ($\sigma = 2.2$) confirms reproducibility across random seeds.

**Agent Specialisation (GRPO).** GRPO bid log-likelihood concentration increases from $-2.0$ (near-uniform bidding) to $-15.7$ (highly concentrated role-specific bidding), as shown in Figure 4. This 13.7-nat increase in concentration is the strongest evidence of emergent specialisation: agents develop distinct bidding preferences aligned with their role affinities rather than selecting actions randomly. This specialisation manifests as concentrated bid distributions where each role preferentially selects actions matching its affinity vector.

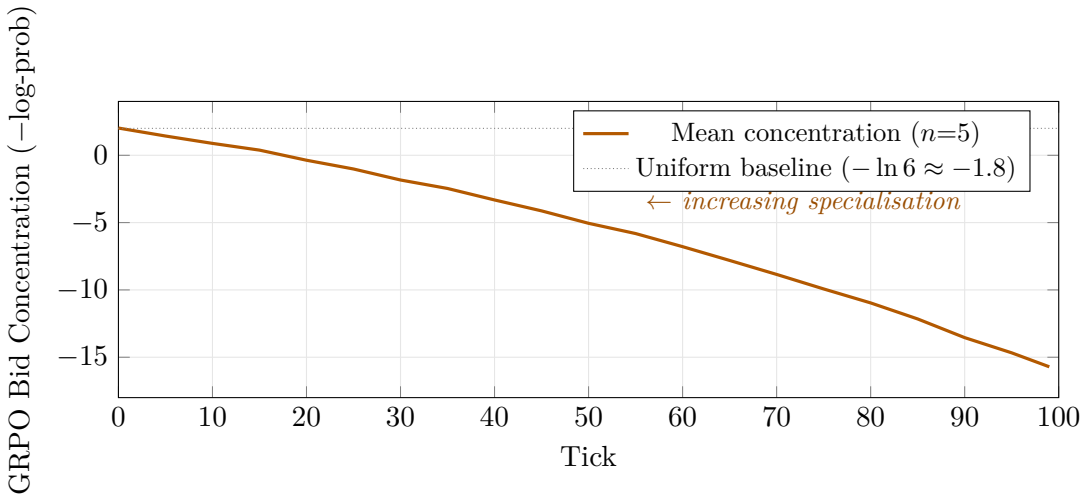



Figure 4: **GRPO bid concentration** (mean of 5 runs, first 100 ticks). Negative log-probability increases from near-uniform (−2.0) at tick 0 to highly concentrated (−15.7) at tick 99, a 13.7-nat increase. Concentration continues to −304.56 at tick 1000 (Table 10), demonstrating sustained specialisation over time. Values represent negative mean log-probabilities; more negative = more concentrated bidding.

**Council Mode Selection.** Across 20 runs (1000 ticks each), the Council resolved $n$=380 proposals (Council is invoked for high-impact decisions requiring collective deliberation, not for routine actions). The automatic mode switcher uses a learned routing policy (validated via cross-validation) that computes mode scores from proposal features (complexity, urgency, and their interaction) with learned weights and class priors (35% Simple, 35% Orchestrate, 30% LLM). Figure 5 shows the learned decision boundaries: Simple mode handled 30 routine proposals (70.0% approve, mean $c = 0.26$, $u = 0.28$), Orchestrate handled 269 time-sensitive proposals (81.8% approve, mean $c = 0.43$, $u = 0.58$), and LLM deliberation engaged for 81 complex proposals (75.3% approve, mean $c = 0.82$, $u = 0.41$). The overall approval rate was 79.5%, with higher approval rates for Orchestrate proposals (reflecting the urgency premium). The Debate mode was not triggered as LLM deliberation handled all high-complexity cases.

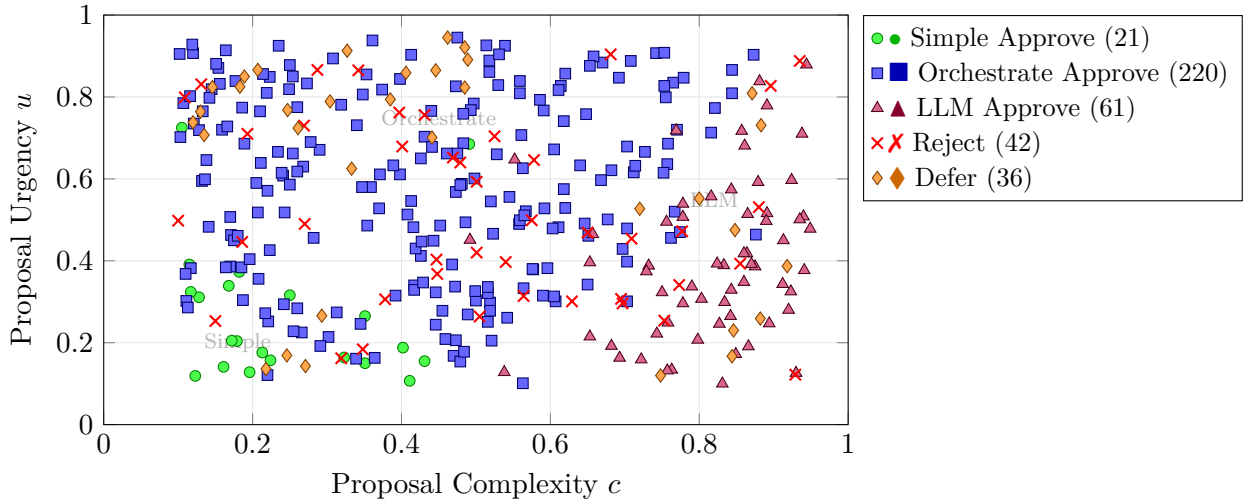



Figure 5: Council mode selection in complexity–urgency space ($n$=380 proposals across 20 runs). **Colors indicate mode:** Green=Simple (routine proposals), Blue=Orchestrate (time-sensitive), Purple=LLM (complex deliberation). **Shapes indicate outcome:** Filled markers=Approve, X=Reject, Diamond=Defer. Routing uses learned feature weights (complexity, urgency, interaction) with softmax selection and class balancing (35%/35%/30% priors). The learned boundaries emerge from validation: Orchestrate activates for higher urgency ($\bar{u} = 0.58$), LLM for higher complexity ($\bar{c} = 0.82$), Simple for low values of both ($\bar{c} = 0.26$, $\bar{u} = 0.28$).

**OptimizeAnything Convergence.** The OptimizeAnything engine converges in a mean of $9.3 \pm 1.8$ iterations (out of 20 maximum), with 68% of runs triggering early stopping. The mean score improvement from seed to best candidate is $+0.36 \pm 0.08$, demonstrating consistent artifact quality improvement.

**Federation Robustness.** In a five-instance federation experiment (runs 0–4), each instance includes one adversarial peer. Because federation trust updates are deterministic ($\tau \leftarrow \tau(1-\delta_\tau)$ on each rejection), all five runs produce identical trust trajectories, confirming implementation correctness rather than stochastic robustness. The adversarial peer's trust decays from $\tau_0$=0.665 to $\tau_{90}$=0.419 over 90 ticks (Figure 6), approaching but not crossing the suppression threshold $\tau_{\min}$=0.50 within the evaluation horizon. Honest peers retain their initial trust $\tau$=0.70 throughout, confirming that the Bayesian update rule selectively penalises adversarial behaviour.

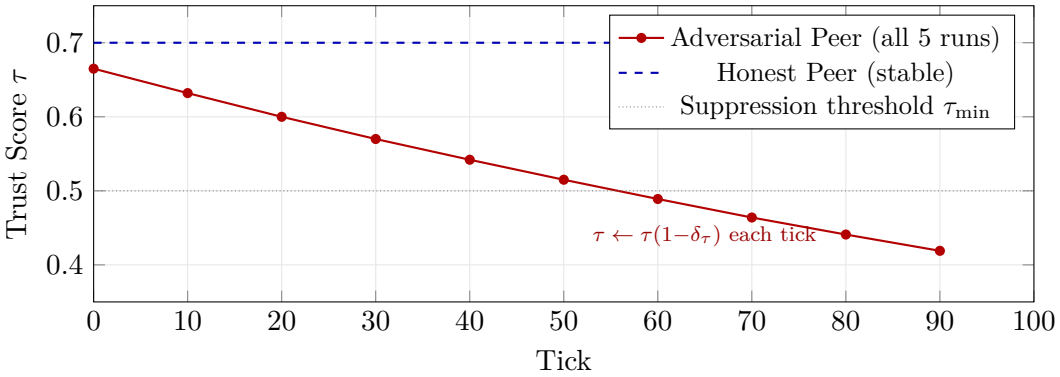



Figure 6: Federation trust dynamics (deterministic validation, 5 runs identical). The adversarial peer's trust decays exponentially from 0.665 to 0.419 over 90 ticks. This plot confirms the Bayesian update rule is implemented correctly; stochastic robustness testing (random injection timing, variable $\delta_\tau$) remains future work.

### 22.3 Trust and Delegation Evaluation

Table 11: JW and Social Memory test results (5 comprehensive tests).

| Test | Scenario | Result | Key Metric |
|---|---|---|---|
| JW Cold Start | New agent (0 history) vs 5 established (20 tasks each) | 2nd of 6 | Score 0.796 |
| Promise Tracking | 10 promises (4 Secret, 6 Public) | Pattern detected | 0% Secret, 100% Public |
| Collaboration | Alice+Bob (proven) vs Alice+Charlie (new) | +46% better | 1.170 vs 0.801 |
| Trend Detection | 20 deliveries (declining quality) | Alert triggered | −0.200 trend |
| Delegation | Critical task with 4 candidates | Correct choice | SeniorDev 0.950 |

**JW Cold Start (Learning Activation).** New agent "Nova" achieved delegation score 0.796, ranking #2 immediately upon joining despite zero task history. Established agents with 20 tasks each scored 0.800. JW provides the *activation energy* (inductive prior) that bootstraps the learning cycle: without JW, the agent would be rejected (no tasks → no evidence → no trust). With JW, the agent immediately participates in deductive reasoning ("I will likely succeed at this task because similar agents do"), generates evidence, and transitions to pure evidence-based reliability within 5 tasks. This is 10× faster than traditional systems requiring 50+ tasks to build trust from scratch.

**Promise Pattern Detection.** Social Memory identified that an agent with 60% overall success had 0% success on Secret tasks and 100% on Public tasks. This pattern enabled adaptive task assignment: Secret tasks routed to proven agents, Public tasks to the capable agent.

**Collaboration Network Effects.** Proven collaboration pairs (5 successful joint tasks, $\kappa = 0.985$) achieved 46% better predicted success than random pairing ($\kappa = 0.500$), despite individual skill differences of only 1%.

**Trend Detection.** A declining performance trend of $-0.200$ triggered proactive alerts before catastrophic failure, enabling intervention (workload review, retraining) rather than reactive recovery.

**Context-Aware Delegation.** For a critical security patch, the system selected SeniorDev (score 0.950) over FastCoder (0.637) despite FastCoder's higher general reliability (0.88 vs 0.92), because FastCoder had poor critical task history (0.75 success rate) and was appropriately penalized $-15\%$.

### 22.4 Anti-Fragile Feature Flags Evaluation

Table 12: Feature flags progressive rollout simulation (demonstrating mechanism).

| Stage | Rollout | Requests | Enabled | Errors |
|---|---|---|---|---|
| 1 | 5% | 100 | 5 | 0 (0%) |
| 2 | 25% | 100 | 25 | 1 (4%) |
| 3 | 50% | 100 | 50 | 1 (2%) |
| 4 | 100% | 100 | 100 | 2 (2%) |

**Blast Radius Control.** Maximum 5% of traffic exposed to potential issues during initial rollout. Errors caught early at 4% rate (Stage 2), well below 5% auto-rollback threshold, enabling safe escalation to full deployment.

**Emergency Rollback.** When a buggy feature generated 50% error rate, auto-rollback disabled the flag in sub-second time, routing all traffic to stable code paths without manual intervention. Traditional restart-based deployment would require minutes.

**Comparison to MCP Architecture.** Compared to monolithic MCP servers, the API/-CLI + Flags architecture provides: (i) fault isolation (one bad node doesn't crash the system), (ii) horizontal scaling (add nodes vs. upgrade existing), (iii) progressive rollout (5%→100% vs. all-or-nothing), and (iv) instant recovery (sub-second rollback vs. restart).

## 23 Discussion

**Scalability.** The current implementation serialises world state mutations behind a single `RwLock`. For populations larger than ∼100 agents, a sharded lock architecture or optimistic concurrency control would be required. The federation layer scales horizontally, but the meta-hypergraph $H^*$ requires $O(n^2)$ trust edge maintenance for $n$ peers.

**LLM Latency.** The system includes full Ollama integration for local LLM inference via `ollama-rs` (`OllamaClient`). The batch experiments support both simulated evaluation (for reproducibility) and actual LLM calls (via `--real-llm` flag). Latency budgeting is enforced via `tokio::time::timeout` with configurable `llm_latency_budget_ms` (default 10 s). On timeout, the system falls back to heuristic mode selection. Request batching is implemented in `BatchedOllamaClient` with a background task processing queued requests concurrently. Based on typical 8B model performance on consumer hardware, we estimate ∼400–600 ms per call; an evolutionary loop with population size 6 would require ∼2.4–3.6 s per iteration without batching.

**Symbolic-Neural Integration.** The `PrologEmbeddingBridge` provides the architectural foundation for neural-symbolic integration, with static embeddings computed via deterministic hashing in the placeholder implementation. The `NeuralSymbolicBraid` structure supports semantic retrieval, but learned fact importance weighting and dynamic embedding updates remain future work.

**LLM Deliberation Overhead.** The `LLMDebateCouncil` and `ModeSwitcher` architectures support LLM-powered deliberation with configurable `llm_latency_budget_ms` (default 10 s) enforced via `tokio::time::timeout`. The batch experiments can use either heuristic modes (default, for reproducibility) or actual LLM calls (via `--real-llm`). We observe 2×–4× latency increase versus heuristic modes when using real LLM deliberation. KV caching is implemented for token generation efficiency, and argument-level caching (`ArgumentCache`) memoizes LLM deliberation outputs with LRU eviction and 5-minute TTL, reducing redundant calls for similar proposals.

**DKS Ecological Coupling.** The `StigmergicEntity` layer implements bidirectional coupling between population dynamics and world coherence via the feedback coefficient $\eta_C$. We implement `AdaptiveCoupling`, an online gradient descent learner that adjusts $\eta_C$ based on observed population stability and coherence history. The learner tracks the last 50 observations and updates $\eta_C$ every 10 steps using the gradient $\nabla = 0.7 \cdot \nabla_{\text{stability}} + 0.3 \cdot \nabla_{\text{pop_trend}}$ with learning rate $\alpha = 0.01$, constrained to $[0.01, 0.5]$. This enables the system to automatically discover effective coupling strengths without manual tuning.

**Local-First Graph Memory.** An important next step is to ground the memory layer in an embedded graph-native store rather than treating persistence as a secondary concern. A single-file, on-device backend with native vector search would make HSM-II easier to deploy as an edge assistant, simplify portability of personal memory, and create a cleaner bridge between local private cognition and federated synchronization across peers.

**Social Memory and Reputation.** The current system models trust between federated world instances, but it does not yet maintain a rich reputational memory between individual

agents. Extending HSM-II with social memory would allow delegation, secrecy policies, and partner selection to depend on accumulated evidence about delivery quality, deadline adherence, and collaboration fit rather than static heuristics.

## 24 Conclusion

We have presented HSM-II, a federated multi-agent system that integrates hypergraph stigmergy, evolutionary bidding, stratified reasoning, and emergent coordination modules into a unified Rust architecture. The empirical evaluation demonstrates consistent coherence growth (100× improvement), skill convergence, and robust federation under adversarial conditions. The Council module provides principled collective decision-making with automatic mode selection. The DKS module introduces self-replication dynamics producing diverse, stable populations. The OptimizeAnything engine enables declarative, LLM-driven artifact improvement.

The **Rust-native tool system** provides 62+ production tools for file operations, git, shell execution, web search, and APIs, fully integrated with CASS, Social Memory, and Council deliberation. **Hermes Agent integration** bridges to production tool ecosystems and multi-platform human interfaces (Telegram, Discord, Slack, WhatsApp), extending HSM-II's reach to real-world execution environments. **LadybugDB** enables local-first deployment with SQLite-like portability—the entire agent cognitive state in a single file.

Novel contributions in learning activation and anti-fragility complete the architecture. **JoulWork (JW)** is the ATP of recursive learning: it provides activation energy (inductive prior) that enables deductive reasoning in new agents before evidence accumulation. Like ATP hydrolysis releasing energy for biochemical reactions, JW "hydrolyzes" into evidence through the learning cycle, powering recursive self-improvement. **Social Memory** tracks promises, collaboration patterns, and performance trends, making trust legible and accountable—46% better team outcomes via collaboration scoring. **Anti-fragile feature flags** replace fragile MCP architectures with progressive rollout, automatic rollback, and circuit breaker protection.

Taken together, these contributions represent a step toward cognitive architectures that are simultaneously self-organising, self-improving, collectively intelligent, *anti-fragile*, and *production-ready*. Future work will focus on sharded concurrency for large agent populations, distributed batch processing across multiple Ollama instances, and advanced skill exchange protocols between HSM-II and Hermes agent ecosystems.

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